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Designing a Secure, Governed and Scalable GenAI Capability

A Phased Roadmap to a Hybrid Operating Model

February 2026

Executive Summary

- **This document was developed as an independent exercise to outline how regulated enterprises can design, govern and scale a controlled GenAI capability.**
- Most organizations are currently in a fragmented experimentation phase, with isolated pilots, standalone tools and limited visibility over risks, costs and model performance.
- Scaling GenAI requires treating it as a controlled enterprise capability, supported by shared platforms, standardized architecture patterns and clear governance.
- GenAI use cases vary significantly in automation level and risk profile (assistive, workflow-embedded, agentic), requiring differentiated controls and oversight.
- A platform-led federated operating model enables scale by combining central infrastructure and guardrails with domain-led use case delivery.
- Enterprise adoption typically evolves through three stages: experimentation → controlled deployment → enterprise-scale integration into core workflows.
- When industrialized correctly, GenAI can deliver measurable productivity gains, improved decision support and operational efficiency while remaining compliant with enterprise risk frameworks.

Agenda

1. How to Frame GenAI as a Controlled Enterprise Capability?
2. Why GenAI must be industrialized?
3. What enterprise-grade looks like?
4. What Operating Model Enables GenAI to Scale Safely Across the Bank?
5. How Do We Transition from Today's Experiments to an Enterprise GenAI Capability?

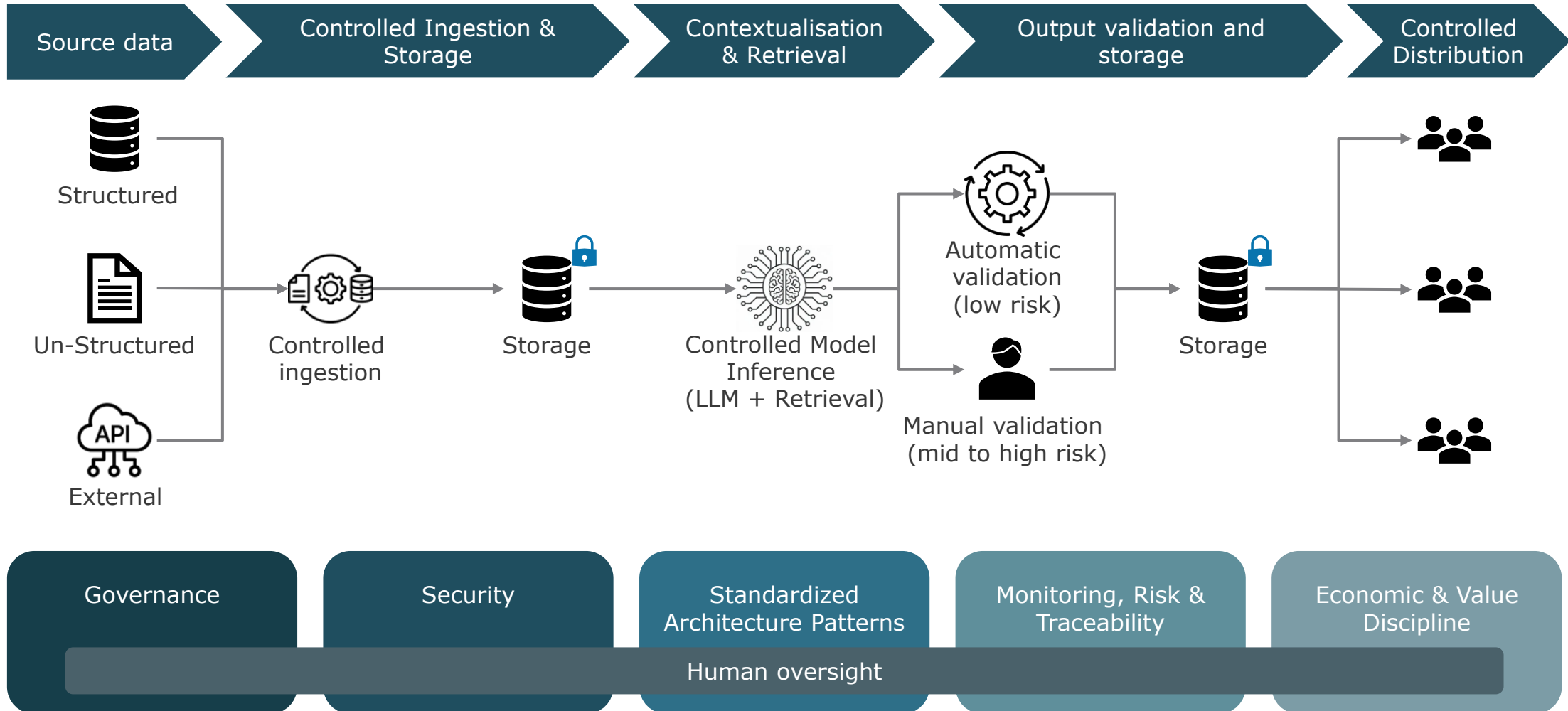
1 – How to Frame GenAI as a Controlled Enterprise Capability?

GenAI Use Cases Differ in Nature and Require Appropriate Levels of Control

	Assistive GenAI ¹	Workflow-Embedded GenAI	Autonomous / Agentic GenAI
Examples	• Drafting communications • Summarizing credit papers and policies • Preparing reports and presentations • Code generation and review support • Extracting insights from meeting notes	• Auto-generating credit memos within workflow • Populating structured fields in core systems • Recommending case routing in case tools • Pre-filling regulatory reports from internal data	• Automatically resolving low-risk cases • Triggering predefined client communications • Executing remediation workflows end-to-end • Updating multiple systems within a workflow • Monitoring portfolios and initiating threshold actions
Traits	• Human initiates • Human reviews • No system action • Advisory output only	• Integrated into workflows • Generates structured outputs • Supports structured decision processes • Still human approval before action	• Multi-step reasoning • System-to-system interaction • Can trigger actions • Limited human gating • Potential closed-loop automation
Risks	Low to moderate	Moderate to high	
			Focus of this document

1: restricted to authorized platforms, with activity logging, information security compliance and no external data sharing

GenAI is an extension of the traditional enterprise data workflow



2 – Why Industrialise GenAI Now?

Failing to Industrialize GenAI Creates Structural Security, Productivity and Innovation Gaps

What We See Today

GenAI activity across the bank is currently driven by local experimentation and assistive use cases.

- Multiple **small GenAI pilots across teams and functions**
- Focus on **assistive productivity tools** (summarization, drafting, search)
- Direct access to models through **standalone tools or lightweight integrations**
- Use cases developed **independently within teams**
- Limited reuse of **prompts, architectures or components**
- GenAI capabilities often **embedded in personal workflows rather than enterprise systems**

Implications for Scaling

While experimentation is valuable, the current model creates structural limitations for scaling

- Duplication of solutions and fragmented tooling across teams
- Limited visibility of **usage, costs and model performance**
- Inconsistent **governance, security and architecture practices**
- Risk of **sensitive data leakage or inappropriate data sharing**
- Increased risk exposure, including **misuse of GenAI tools**
- Growing **regulatory and conduct risk** without central controls

Early experimentation is essential for learning, but without a standardized platform and operating model, GenAI adoption becomes fragmented, costly and difficult to control at scale.

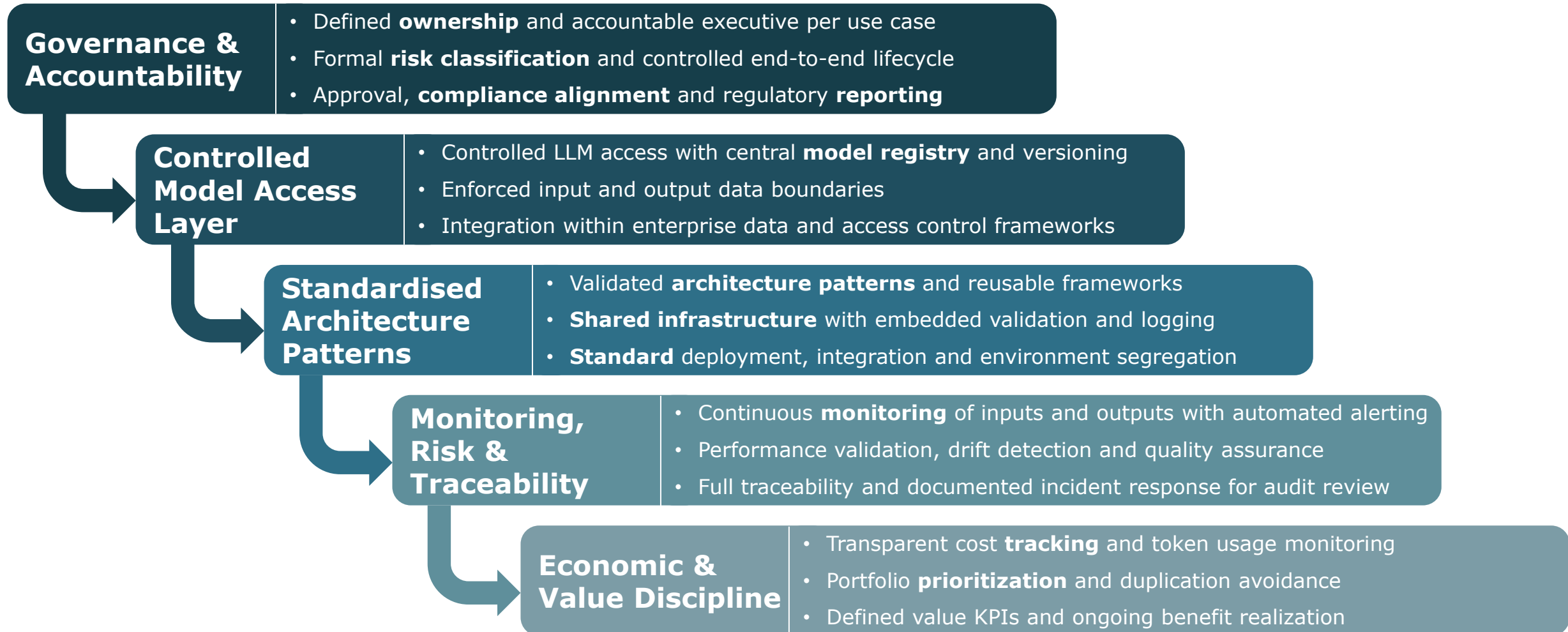
Industrializing GenAI Converts Fragmented Experiments into Scalable Enterprise Value



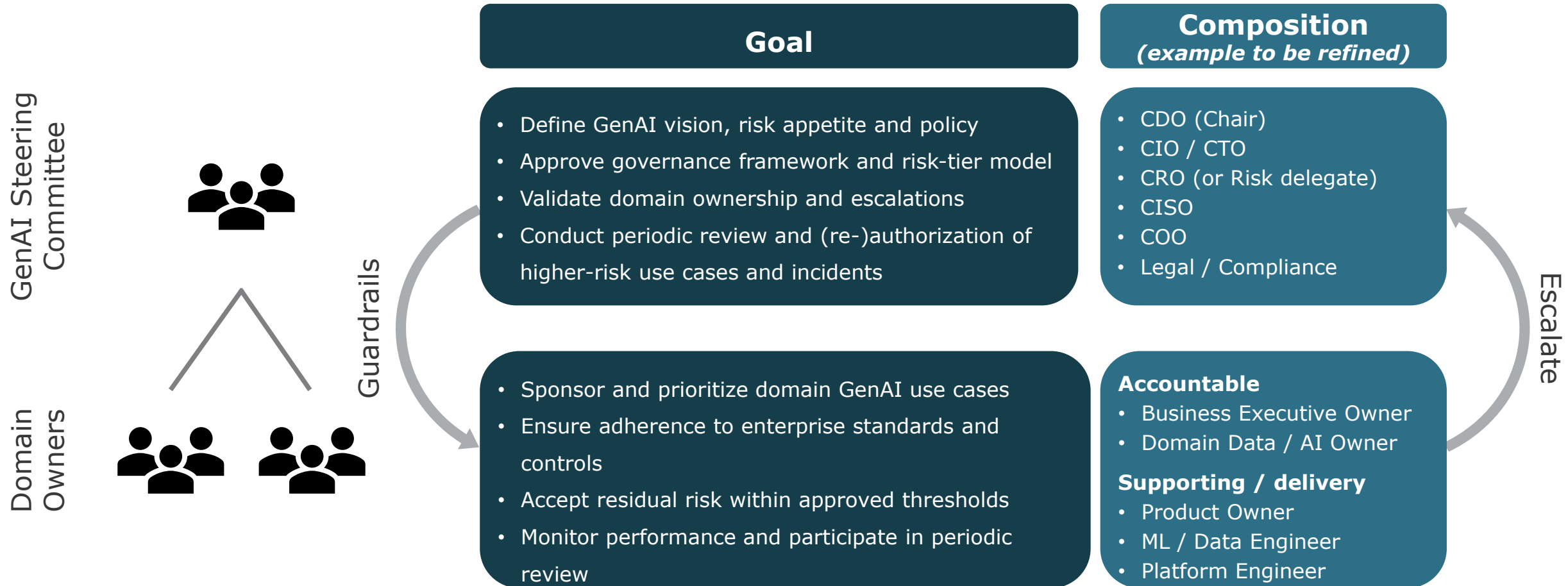
GenAI value scales only when deployed on a controlled enterprise platform

3 – What enterprise-grade looks like?

Scaling GenAI safely requires governance, platform controls and value discipline



Clear Governance and Accountability Enable Enterprise-Wide GenAI Adoption



Model centralization ensures GenAI operates within secure and controlled enterprise boundaries

Controlled Model Access

- Enterprise catalogue of approved models
- All variants centrally registered
- Production subject to risk-based approval
- Direct external API usage prohibited

Data & Prompt Guardrails

- Input filtering aligned to data classification
- Output validation and policy enforcement
- Controlled retention of prompts and responses
- Enterprise data excluded from external model training
- Data classification, handling and retention controls for prompts, context and outputs



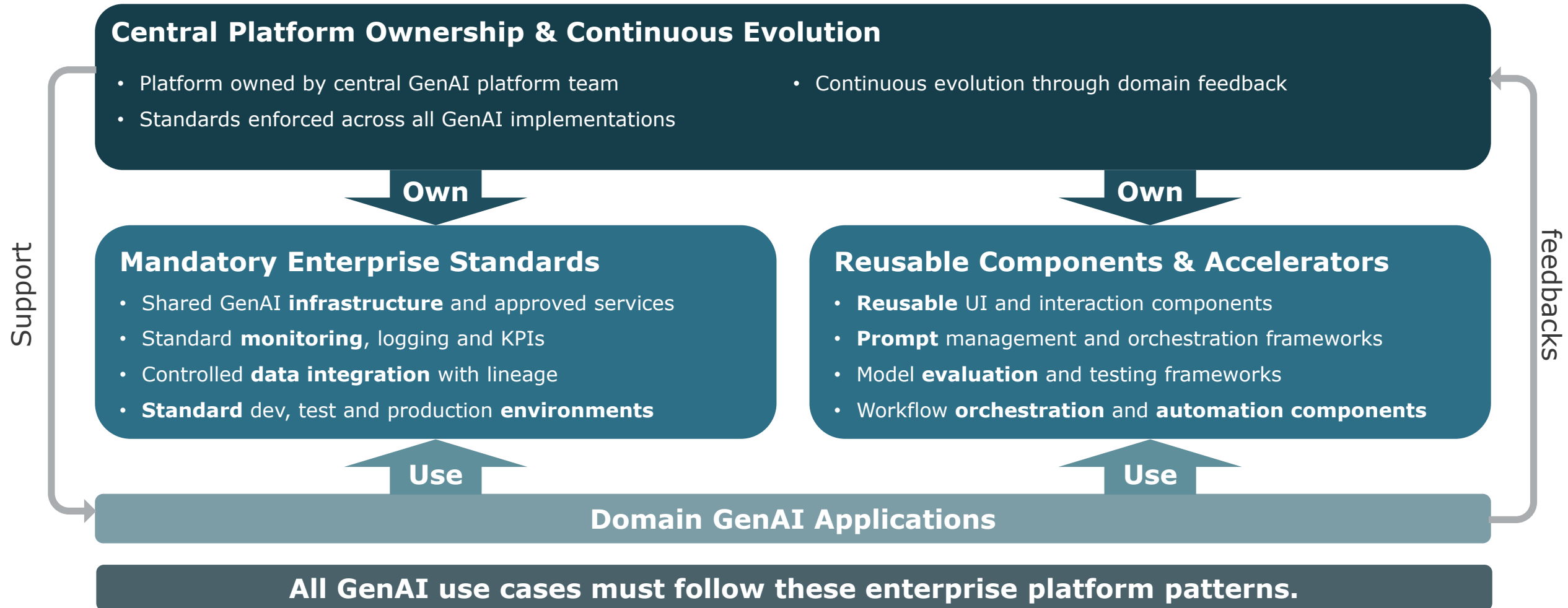
Identity & Access Control

- Access governed through enterprise IAM
- Role-based entitlement to model endpoints
- Strict dev, test, production segregation
- Privileged administrative access monitored and restricted
- Protection against prompt injection, adversarial inputs and model misuse

Traceability & Audit

- Immutable logging of all model executions
- User, model and version traceability
- End-to-end execution traceability
- Audit-ready evidence for regulatory review

Standardized GenAI Architecture Patterns Enable Controlled and Scalable Delivery



Continuous Monitoring and Traceability Enable Effective GenAI Risk Management

Traceability – *Capturing full execution evidence*

- Logging of prompts, responses and model versions
- User identity and execution context recorded
- End-to-end traceability of model execution
- Audit-ready execution history

Enables

Monitoring – *Continuous supervision of model behavior*

- Performance monitoring and model drift detection
- Output quality, hallucination and policy compliance checks
- Usage analytics and anomaly detection
- Automated dashboards and alerting

Manage

Risk – *Managing Model and Operational Risks*

- Detection of hallucinations, bias, misuse, adversarial attacks and business impact risks
- Investigation of abnormal model behavior
- Incident escalation and response procedures
- Continuous improvement of controls

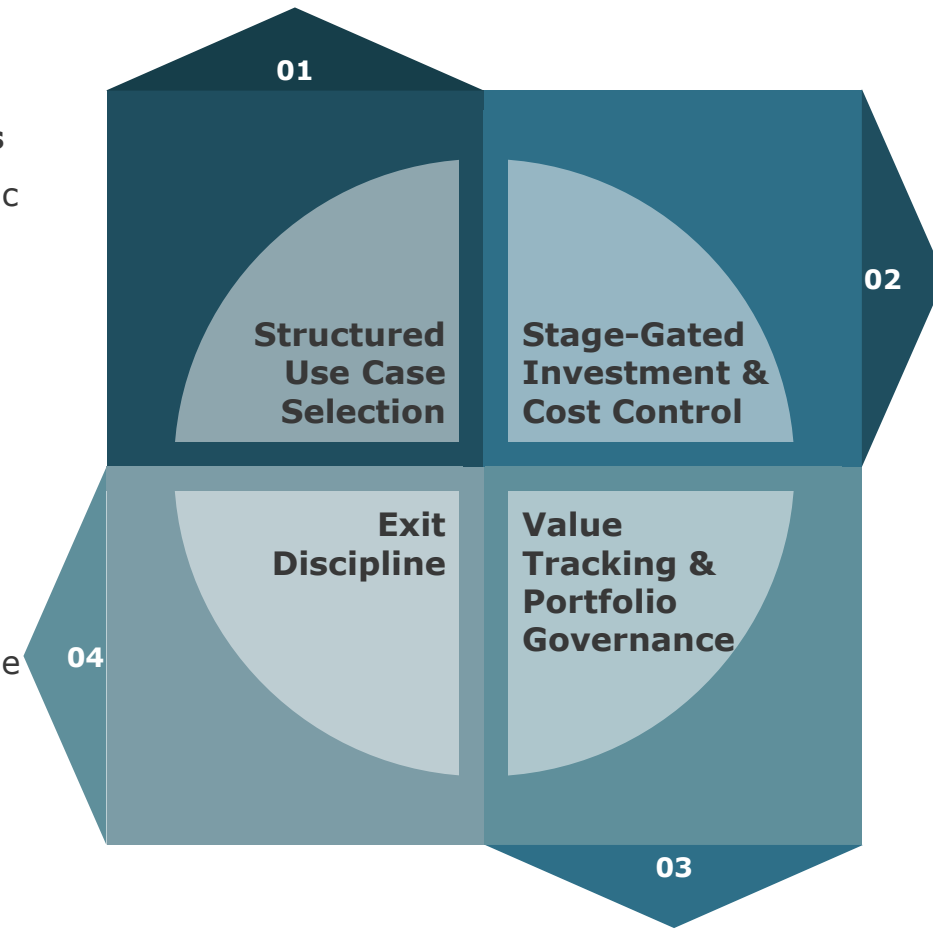
Economic Discipline Ensures GenAI Investments Deliver Measurable Value

Prioritizing Highest-Value Opportunities

- Prioritized by business impact and strategic alignment
- Assessed for cost, risk and data
- Approved through GenAI forum

Stopping initiatives that do not deliver

- Reviews at defined stage gates
- Stop initiatives without demonstrated value
- Reallocate capacity to higher-value use cases



Funding aligned to validated value

- POC funding to test feasibility
- Pilot funding for controlled deployment
- Scale funding after proven value

Ensuring measurable outcomes

- Standard dashboards track usage and costs
- Global KPIs¹ applied across use cases
- Business sponsors accountable for value

1: Examples of KPIs tracked: Number of active users, usage frequency, Cost per request / generated output, Model latency and reliability, Time saved in supported workflows, Reduction in manual processing effort, Override or correction rate of outputs

Aligning GenAI with Existing Risk and Resilience Frameworks Ensures Regulatory Compliance

GenAI Model Lifecycle Governance

Aligning GenAI with the bank's model risk framework

- Risk classification aligned with model risk policy
- Independent validation for higher-risk models prior to production use
- Central model registry and version control
- Performance monitoring and periodic review
- Formal change management and retirement process

Third-Party & Model Vendor Risk

Managing external model providers within outsourcing controls

- Approved vendor catalogue and due diligence
- Legal safeguards on data usage and training rights
- Monitoring vendor performance and dependency risks
- Multi-model strategy to reduce concentration risk
- Defined exit and model substitution strategy

Data Sovereignty & Operational Resilience

Ensuring secure, resilient and compliant GenAI operations

- Enforcement of data residency and storage policies
- Secure handling of prompts, context data and outputs
- Strict dev / test / production environment segregation
- Human override and controlled model fallback mechanisms
- Full execution logging and audit traceability

These controls align GenAI with the risk and resilience frameworks expected in regulated banking.

4 – What Operating Model Enables GenAI to Scale Safely Across the Bank?

A Platform-Led Federated Operating Model Enables Safe Enterprise GenAI Scaling

Central GenAI Platform – *Providing shared infrastructure and guardrails*

Platform Infrastructure

- GenAI platform
- Cloud infrastructure
- LLM gateway & model access management
- Central model registry and versioning

Shared Capabilities

- RAG services
- Vector databases and embedding services
- Reusable GenAI frameworks¹
- Prompt libraries and tooling

Operational Services

- Monitoring and logging
- Security controls and access management
- Usage analytics and cost tracking

Guardrails

Enable

Escalate

Domain Use Case Squads – *Delivering business value through GenAI*

Use Case Delivery

- Identifying and prioritizing GenAI opportunities
- Building domain GenAI applications and workflows

Domain Integration

- Integrating domain data products
- Contextualizing knowledge through RAG

Adoption & Value creation

- Embedding GenAI in operational processes
- Tracking business outcomes and productivity gains

Oversight

Risk & Governance Functions

Ensuring GenAI operates within the bank's control framework

Model Governance

- Model risk classification and validation
- Lifecycle approval and review
- Model monitoring and re-validation

Security & Architecture Compliance

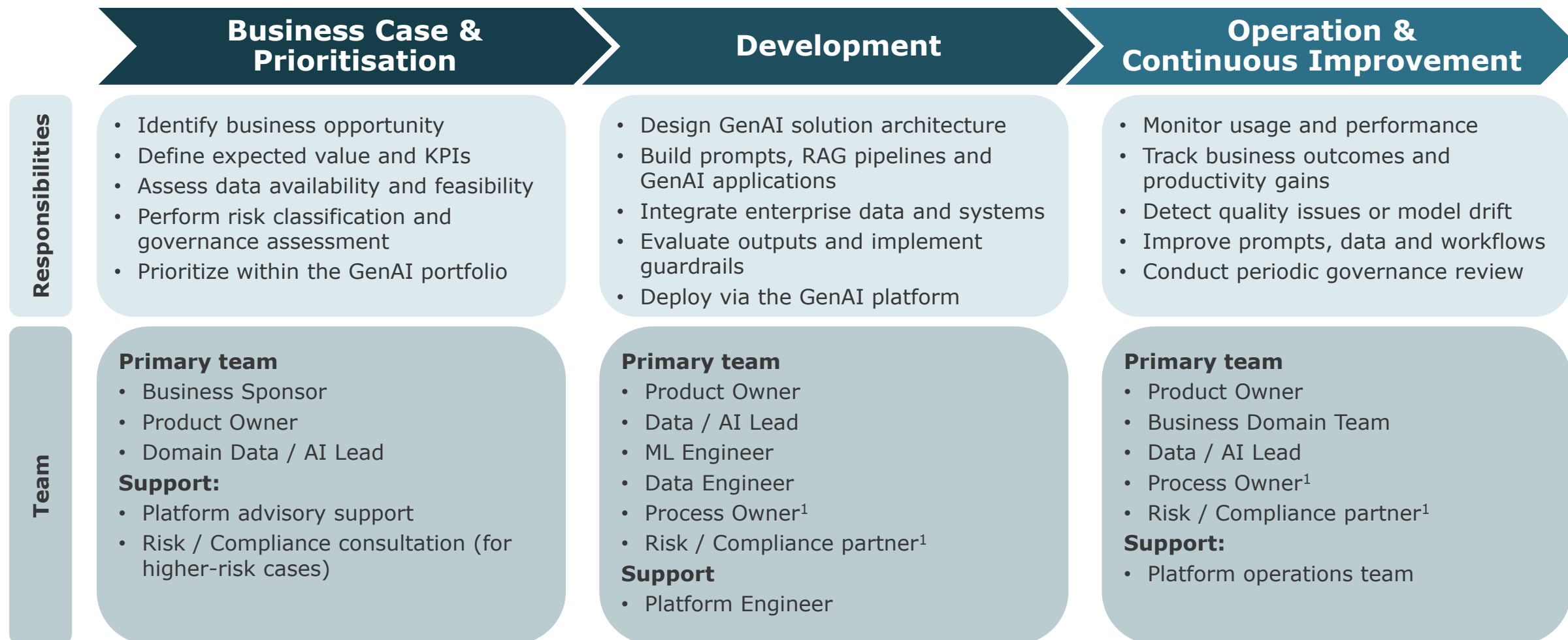
- Data protection and access policies
- Regulatory compliance oversight

Operational Control

- Incident management and escalation
- Monitoring adherence to platform standards

¹: Including evaluation and benchmarking frameworks

Teams evolve as GenAI use cases move from experimentation to production



¹: Applicable for workflow-embedded or autonomous use cases

Human Adoption and Change Management Are Critical to Safe GenAI Deployment

Adoption & Capability Building

- Training users on appropriate GenAI usage, limitations and verification practices
- Developing AI literacy across business teams, including prompt design and output validation
- Establishing clear usage policies and guidelines for enterprise GenAI tools
- Embedding GenAI into workflows with defined human accountability and oversight

Managing Behavioural and Operational Risks

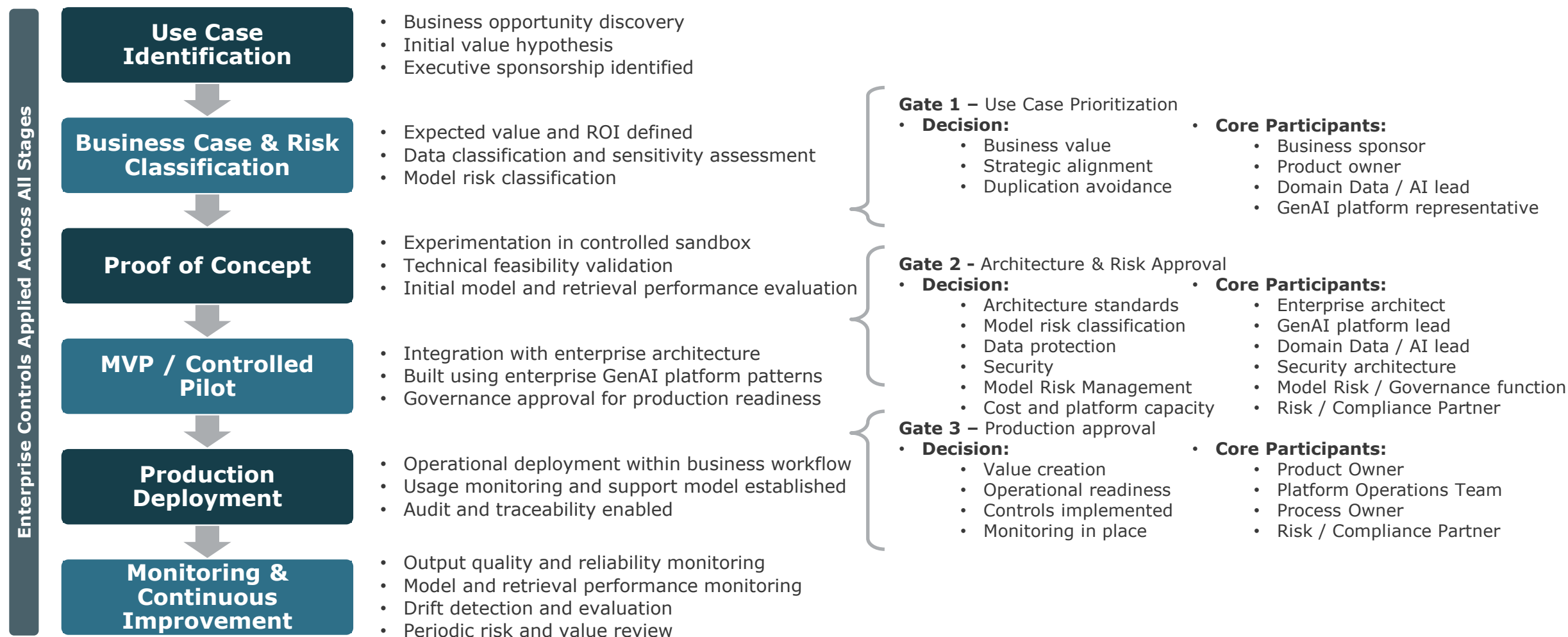
- Preventing automation bias and over-trust in AI-generated outputs
- Reinforcing human verification for high-risk decisions or client-facing actions
- Ensuring users understand model limitations, uncertainty and hallucination risks
- Monitoring usage patterns and override rates to detect excessive reliance on AI

Organizational Change Management

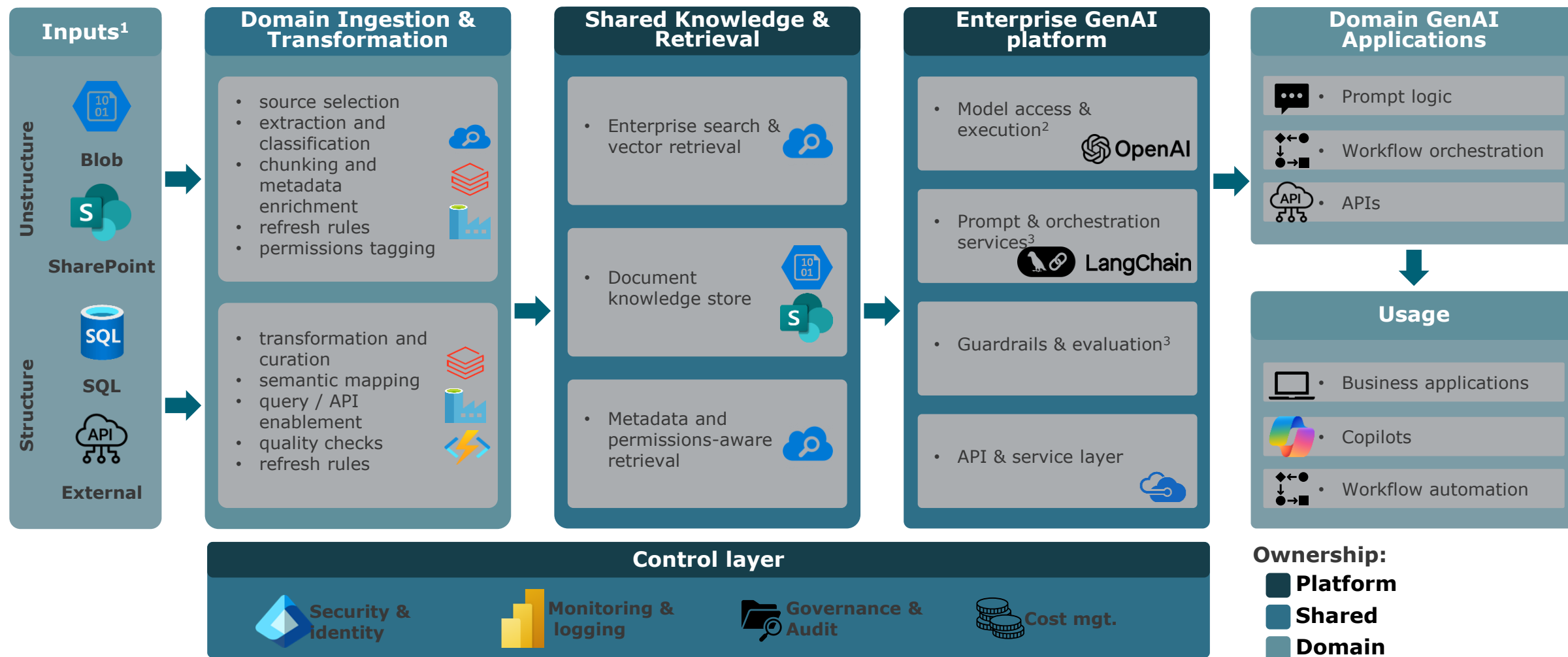
- Defining clear ownership for GenAI-enabled processes
- Updating operating procedures and internal controls when AI is introduced
- Collecting user feedback and operational metrics to refine prompts, workflows and guardrails
- Communicating acceptable use and escalation procedures for AI failures

Safe GenAI adoption depends as much on user behavior and governance as on technology controls.

GenAI initiatives follow a controlled stage-gated lifecycle aligned with enterprise governance



A Central GenAI Platform Provides Shared Infrastructure While Domains Build Business Applications



1: Example to be refined

2: generate responses, create embeddings for retrieval, and route requests to the appropriate model

3: Prompts are standardized, enriched with retrieved context, and orchestrated with tools or external data.

4: Safety filters, validation, and evaluation monitor outputs and detect unreliable responses.

Separating the Central GenAI Platform from Domain Applications Enables Scalable Enterprise Adoption

Central GenAI Platform

Infra. & Model Access

- Azure OpenAI model access
- Model routing and version management
- Embedding generation

Retrieval Infrastructure

- Azure AI Search
- Vector indexes
- Enterprise document retrieval

Orchestration Services

- Prompt management
- RAG orchestration
- Tool / function calling

Guardrails & Monitoring

- Safety filters¹
- Output validation²
- Evaluation frameworks³

Platform Services

- API gateway
- Usage monitoring
- Cost tracking

Security & Governance

- Identity & access mgt.
- Audit logging
- Compliance controls
- Network isolation

Domain GenAI Use Case Architecture

Domain Knowledge & Data

- Domain knowledge sources
- Domain data products
- Domain ingestion pipelines

Integration

- Integration with domain systems
- Business process automation
- APIs to internal services

Application Logic

- Prompt logic
- Domain-specific workflows
- Tasks orchestration

User Interfaces

- Copilots
- Internal applications
- Workflow tools

Quality Measurement

- Accuracy & grounding
- Retrieval relevance
- Safety & reliability

Value Measurement

- Business KPIs
- Productivity metrics
- Adoption tracking

The central platform provides shared infrastructure, models and governance, while domain teams build business-specific applications using these capabilities.

1: e.g. Azure OpenAI content filters, custom policy rules, prompt filtering before the model

2; e.g. Validate outputs against retrieved context or policy rules

3 :e.g. use a golden dataset / human evolution

Enterprise GenAI capabilities rely on five foundational technology layers

Foundation Model Layer

Purpose

- Natural language understanding and generation
- Creating embeddings for retrieval pipelines
- Supporting reasoning and tool/function calling

Examples

- Azure OpenAI (GPT-4 / GPT-4o)
- Anthropic Claude
- Open-source models (Llama, Mistral)

AI Orchestration

Purpose

- Orchestrating multi-step GenAI workflows
- Managing prompts and context
- Integrating tools, APIs, and enterprise systems

Examples

- Semantic Kernel
- LangChain
- Azure AI Studio orchestration capabilities

Enterprise & Cloud-Native Platform

Purpose

- Running GenAI workloads securely and at scale
- Exposing GenAI capabilities through APIs
- Integrating with systems and workflows

Examples

- Azure OpenAI Service
- Azure Kubernetes Service (AKS)
- Azure API Management

Knowledge & Retrieval Infrastructure

Purpose

- Storing embeddings of enterprise knowledge
- Retrieving relevant context for user queries
- Enabling Retrieval-Augmented Generation (RAG)

Examples

- Azure AI Search (vector search)
- Pinecone
- Document stores (SharePoint, Blob Storage)

Guardrails, Monitoring & Evaluation

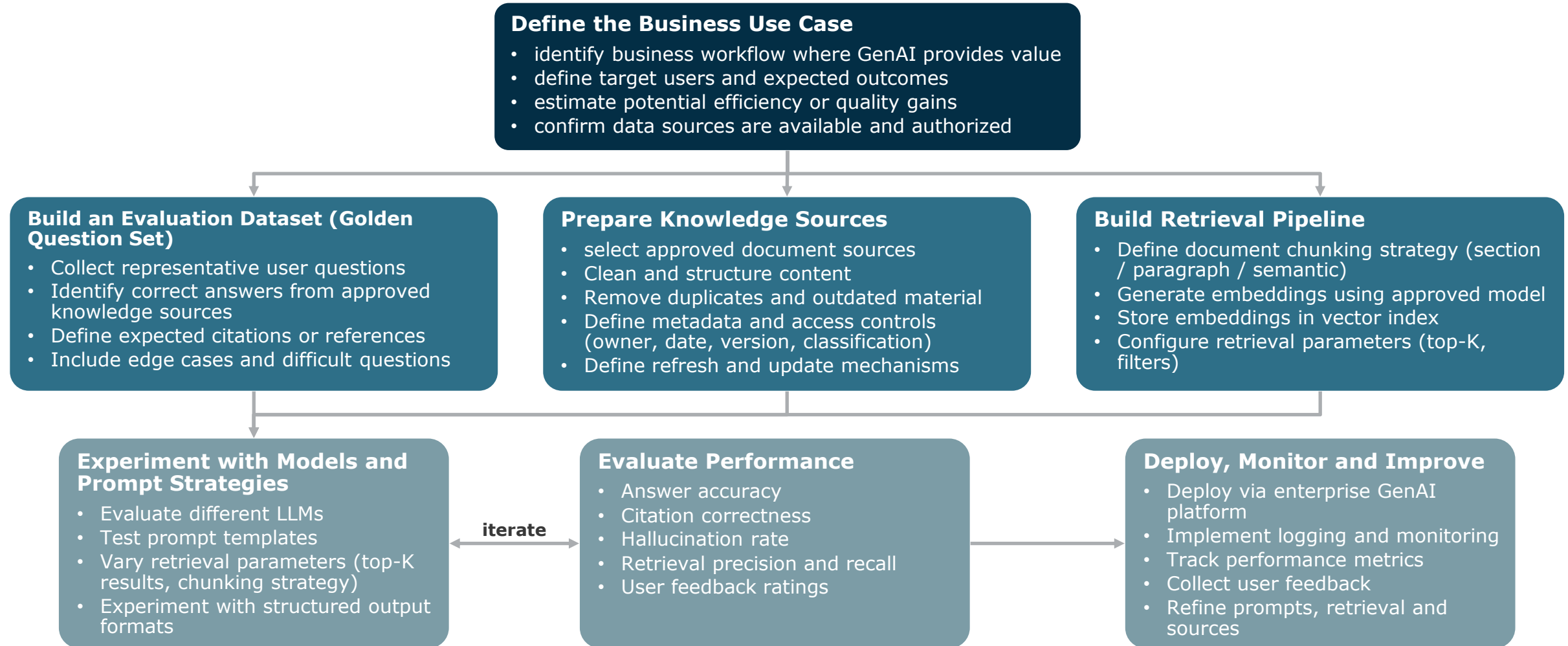
Purpose

- Applying safety and policy filters
- Monitoring model behaviour & performance
- Evaluating response quality
- Detecting hallucinations

Examples

- Azure OpenAI content filters
- Azure AI Content Safety
- Evaluation frameworks (Prompt Flow, DeepEval)
- Monitoring platforms (Azure Monitor, Datadog)

Building Reliable RAG Systems Requires Structured Experimentation and Evaluation



Combining Automated Monitoring and Human Oversight Ensures Reliable and Compliant GenAI Outputs

Answer Quality & Relevance

Objective: *Ensure GenAI responses are accurate, relevant and useful for the intended task*

Evaluation methods

- Human review of sampled responses
- Benchmark test questions
- User feedback and ratings

Example metrics

- Response accuracy
- Relevance to user query
- Hallucination rate

Grounding & Knowledge Accuracy

Objective: *Ensure responses are based on trusted enterprise knowledge rather than model speculation*

Evaluation methods

- Validation against approved knowledge sources
- Citation checks in RAG responses
- Retrieval quality testing

Example metrics

- Retrieval precision and recall
- Percentage of responses with valid citations
- Knowledge coverage of source material

Safety, Compliance & Risk Controls

Objective: *Ensure GenAI outputs comply with regulatory requirements and internal policies*

Evaluation methods

- Prompt safety testing and red teaming
- Sensitive data exposure checks
- Policy compliance validation

Example metrics

- Unsafe output rate
- Policy violation rate
- Sensitive data leakage incidents

Operational Performance

Objective: *Ensure the GenAI system performs reliably and efficiently in production*

Evaluation methods

- Production monitoring
- Usage analytics
- Latency and reliability tracking

Example metrics

- Response latency
- Usage adoption
- Cost per request

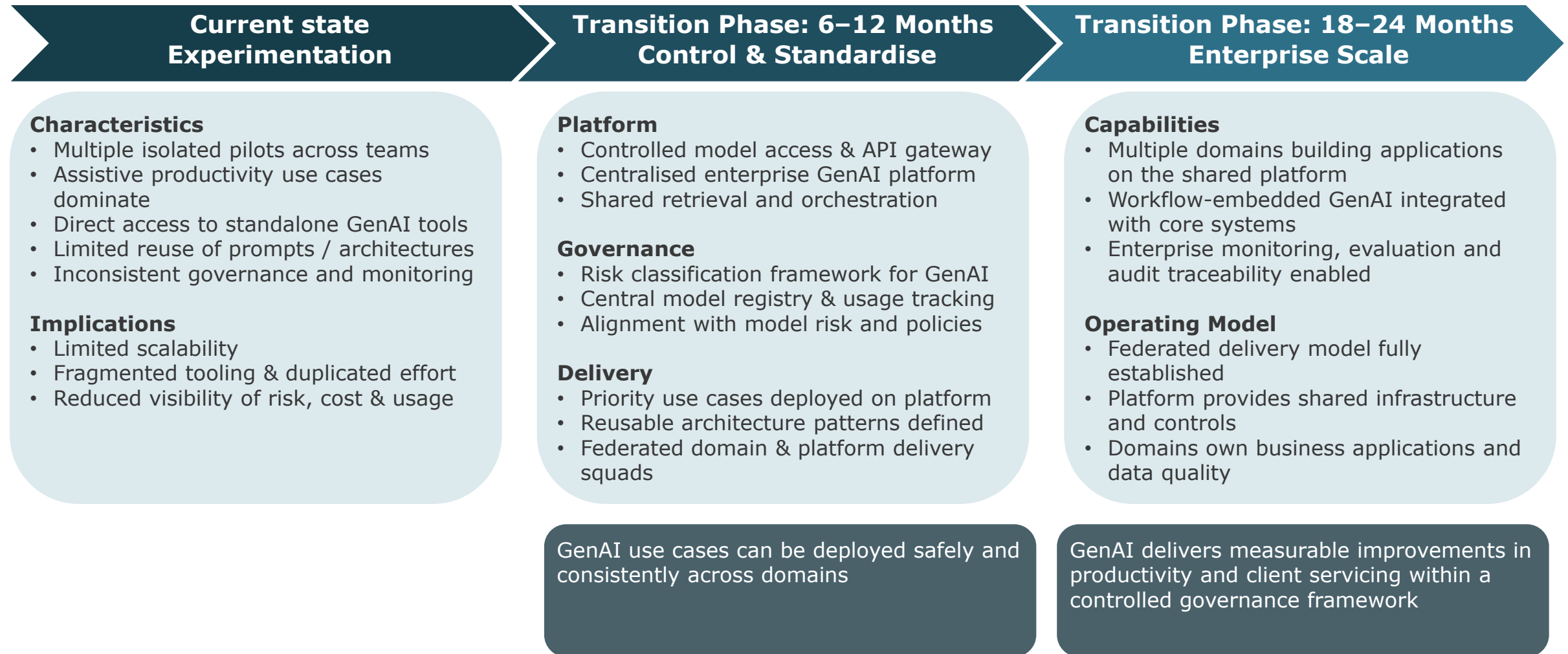
GenAI Costs Increase with Automation Complexity but Decrease with Platform Reuse

Use Case Type	Phase	Typical Activities	Effort ¹ (Man-Days)	Operating Cost ¹
Assistive GenAI	1 – Proof of Concept	Prompt experimentation, initial RAG pipeline, prototype interface, evaluation dataset	40 – 80	–
	2 - MVP / Controlled Deployment	Production RAG pipeline, integration with knowledge sources, access control, monitoring	60 – 120	–
	3 – Production Monitoring	Monitoring, prompt and retrieval tuning, knowledge updates, operational support	5 – 10 / month	£2k – £15k / month
Workflow-Embedded GenAI	1 – Proof of Concept	Workflow mapping, structured output generation, prototype integration	30 – 60	–
	2 - MVP / Controlled Deployment	System integrations, structured output validation, governance approval, monitoring setup	150 – 300	–
	3 – Production Monitoring	Operational monitoring, governance review, integration maintenance	10 – 20 / month	£10k – £50k / month
Agentic / Automation GenAI	1 – Proof of Concept	Agent design, tool orchestration, workflow simulation	40 – 120	–
	2 - MVP / Controlled Deployment	Multi-system integration, workflow automation, safety controls, evaluation	400 – 700	–
	3 – Production Monitoring	Continuous monitoring, incident handling, optimisation	20 – 60 / month	£20k – £100k / month

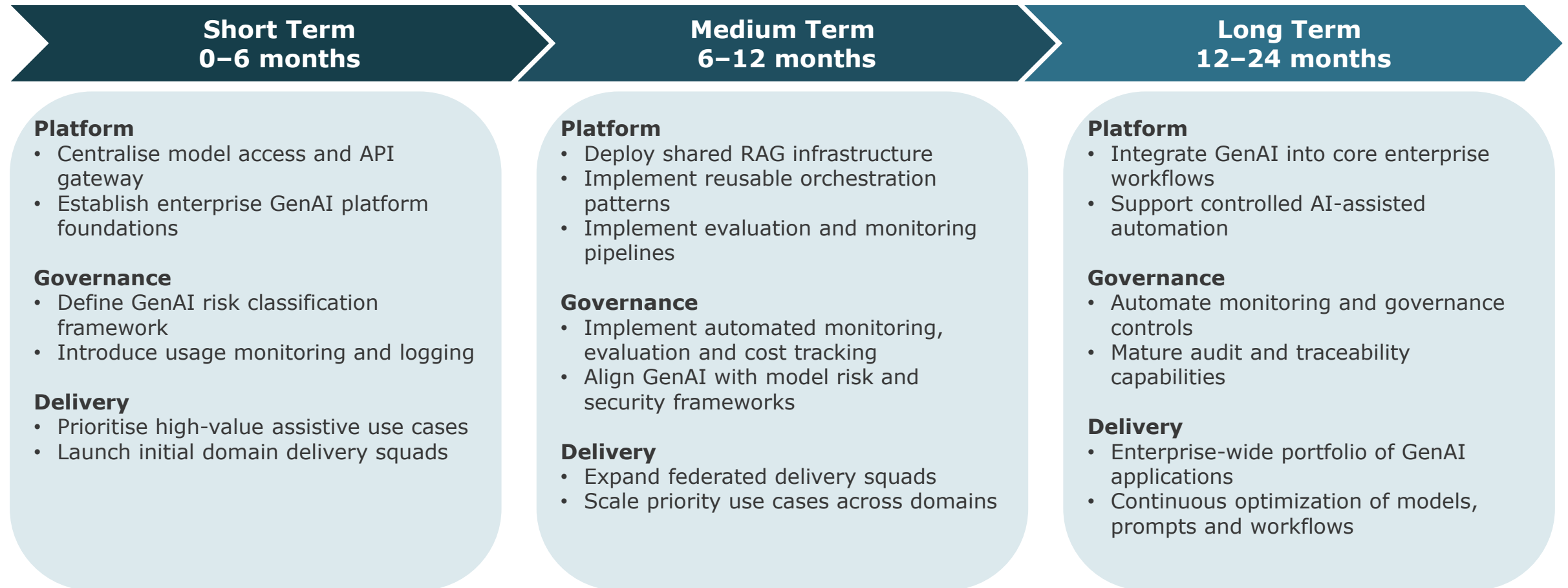
Platform reuse significantly reduces marginal effort and cost once the core GenAI platform and retrieval infrastructure are in place.

5 – How Do We Transition from Today's Experiments to an Enterprise GenAI Capability?

How Do We Transition from Experimentation to an Enterprise GenAI Capability?



What Is the Roadmap to Achieve Mature Enterprise GenAI Capability?





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Head of Data / Data Platforms leader with **15+ years of experience designing and scaling enterprise data capabilities across banking and private equity.**

Combines strategy consulting and hands-on platform delivery to **translate executive priorities into production-grade data platforms, analytics and AI capabilities.**

Key Expertise:

- Enterprise **Data & AI Strategy**
- **Data Platform Architecture** (Azure, Databricks)
- **GenAI industrialisation** and AI governance
- **Data operating models** and delivery frameworks
- **Regulated environments** (banking, financial services, private equity)

Annexes

Role description

Business & Product Roles

Business Sponsor

- Senior business leader accountable for the strategic value of the GenAI use case.
- Approves the business case, sponsors funding, and ensures alignment with business priorities.

Product Owner

- Owns the GenAI use case across its lifecycle.
- Defines requirements, prioritizes features, coordinates delivery teams, and ensures business value is realized.

Process Owner

- Responsible for the operational process where GenAI is embedded.
- Ensures the solution integrates effectively into workflows and that operational controls and oversight are maintained.

Data & AI Roles

Domain Data / AI Lead

- Leads the design of the GenAI solution within the domain.
- Ensures the use case aligns with enterprise data standards, selects appropriate models, and oversees AI solution quality.

ML Engineer

- Develops and integrates machine learning and GenAI components.
- Implements prompt pipelines, model orchestration, evaluation mechanisms, and deployment into production environments.

Data Engineer

- Builds and maintains data pipelines and integrations required for GenAI solutions.
- Ensures reliable access to enterprise data and supports retrieval-augmented generation (RAG) pipelines.

Platform & Engineering Roles

Platform Engineer

- Builds and maintains the central GenAI platform services.
- Manages model gateways, RAG infrastructure, deployment pipelines, and platform scalability.

Platform Operations Team

- Ensures the operational stability and performance of the GenAI platform.
- Monitors usage, reliability, security, and cost efficiency of deployed GenAI services.

Risk & Governance Roles

Risk / Compliance Partner

- Advises delivery teams on regulatory, conduct, and model risk requirements.
- Ensures GenAI use cases operate within the bank's governance and regulatory frameworks.

Model Risk / Governance Function

- Provides independent oversight of GenAI models and systems.
- Conducts risk classification, validation, and periodic review to ensure models remain compliant and reliable.